Chapter 4. Inference in Simple Regression

Sampling distributions, confidence intervals, prediction, inference on correlation, residuals

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Throughout, $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ and fixed x_i with $S_{xx} > 0$.

4.1 Sampling distributions of the estimators

Motivation. To judge a fitted slope we need its sampling law: normal, and t when studentized.

Both estimators are linear in y:

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \sum_i k_i y_i, \qquad k_i = \frac{x_i - \bar x}{S_{xx}}, \qquad \sum k_i = 0, \quad \sum k_i x_i = 1, \quad \sum k_i^2 = \frac{1}{S_{xx}},$$

(using $S_{xy} = \sum (x_i - \bar x) y_i$), and $\hat\beta_0 = \bar y - \hat\beta_1 \bar x$.

Theorem 4.1

(a) $\hat\beta_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$.

(c) $\hat\beta_0 \sim N\Big(\beta_0,\ \sigma^2\big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\big)\Big)$ and $\operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\frac{\bar x \sigma^2}{S_{xx}}$.

Proof. (a) $\hat\beta_1$ is a linear combination of independent normals, with mean $\sum k_i(\beta_0 + \beta_1 x_i) = \beta_1$ and variance $\sigma^2 \sum k_i^2 = \sigma^2/S_{xx}$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = \sum_i \frac{1}{n} k_i \sigma^2 = 0$.

(c) $E\hat\beta_0 = (\beta_0 + \beta_1 \bar x) - \bar x \beta_1 = \beta_0$ and, by (b),

$$\operatorname{Var} \hat\beta_0 = \operatorname{Var} \bar y + \bar x^2 \operatorname{Var} \hat\beta_1 = \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\bar x \operatorname{Var} \hat\beta_1.$$

■

Theorem 4.2

Let $SSE = \sum e_i^2$ and $MSE = SSE/(n-2)$. Then $SSE/\sigma^2 \sim \chi^2(n-2)$, independent of $(\hat\beta_0, \hat\beta_1)$. In particular $E(MSE) = \sigma^2$.

(The case $p = 1$ of Theorem 6.4; directly, use an orthogonal transformation whose first two rows span $\mathbf{1}$ and $\mathbf{x} - \bar x \mathbf{1}$.) Consequently

$$\frac{\hat\beta_1 - \beta_1}{\sqrt{MSE/S_{xx}}} \sim t(n-2), \qquad \frac{\hat\beta_0 - \beta_0}{\sqrt{MSE(1/n + \bar x^2/S_{xx})}} \sim t(n-2).$$

Corollary 4.3 (Intervals and tests for the coefficients)

$100(1-\alpha)\%$ confidence intervals are

$$\beta_1 :\ \hat\beta_1 \pm t_{\alpha/2}(n-2)\sqrt{\frac{MSE}{S_{xx}}}, \qquad \beta_0 :\ \hat\beta_0 \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

The test of $H_0 : \beta_1 = 0$ rejects when $|t_0| > t_{\alpha/2}(n-2)$, $t_0 = \hat\beta_1/\sqrt{MSE/S_{xx}}$. If σ^2 is known, replace MSE by σ^2 and t by z.

Theorem 4.4 (ANOVA F test)

$E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(SSE) = (n-2)\sigma^2$. The statistic $F_0 = MSR/MSE$, $MSR = SSR/1$, satisfies $F_0 = t_0^2$, so the F test with $F_\alpha(1, n-2) = t_{\alpha/2}(n-2)^2$ is the two-sided t test of $\beta_1 = 0$.

Proof. $E(SSR) = S_{xx} E\hat\beta_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Also $SSR/MSE = \hat\beta_1^2 S_{xx}/MSE = t_0^2$. ■

So $E(MSR)/E(MSE) = 1 + \beta_1^2 S_{xx}/\sigma^2$, which is 1 exactly when $\beta_1 = 0$; large F_0 is evidence against H_0.

4.2 Estimating the mean response and predicting

Motivation. Predicting a new response must include its own error, so prediction intervals are wider than mean-response intervals.

At a point x, the mean response $\mu_{y\cdot x} = \beta_0 + \beta_1 x$ is estimated by $\hat y = \hat\beta_0 + \hat\beta_1 x = \bar y + \hat\beta_1(x - \bar x)$.

Theorem 4.5

(a) $\hat y \sim N\Big(\mu_{y\cdot x},\ \sigma^2\Big(\dfrac{1}{n} + \dfrac{(x-\bar x)^2}{S_{xx}}\Big)\Big)$, and a $100(1-\alpha)\%$ confidence interval for $\mu_{y\cdot x}$ is

$$\hat y \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}\Big)}.$$

(b) For a new observation $y_s = \beta_0 + \beta_1 x + \epsilon_s$, independent of the sample, $y_s - \hat y \sim N(0, \sigma^2(1 + \frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}))$, and a $100(1-\alpha)\%$ prediction interval is

$$\hat y \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(1 + \frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}\Big)}.$$

Proof. (a) By Theorem 4.1(b), $\operatorname{Var}\hat y = \operatorname{Var}\bar y + (x-\bar x)^2 \operatorname{Var}\hat\beta_1$. Studentize with MSE, which is independent of $\hat y$.

(b) y_s is independent of $\hat y$, so the variances add. ■

Both are narrowest at $x = \bar x$ and widen quadratically (extrapolation is costly); the prediction interval keeps the irreducible σ^2 as $n \to \infty$.

Example 4.1

For the advertising data of Example 3.1 ($n = 6$, $\bar x = 5.667$, $S_{xx} = 45.33$, $MSE = 1.92/4 = 0.48$, $\hat y = 2.37 + 1.346x$), at $x = 7$: $\hat y = 11.79$, $\frac{1}{6} + \frac{(1.333)^2}{45.33} = 0.206$, and $t_{0.025}(4) = 2.776$. The 95% confidence interval for the mean is $11.79 \pm 2.776\sqrt{0.48 \cdot 0.206} = 11.79 \pm 0.87$, and the prediction interval is $11.79 \pm 2.776\sqrt{0.48 \cdot 1.206} = 11.79 \pm 2.11$.

4.3 Inference on the correlation

When (X, Y) is bivariate normal with correlation ρ, the conditional mean is linear with slope $\beta_1 = \rho\sigma_Y/\sigma_X$, so $\rho = 0 \iff \beta_1 = 0$.

Theorem 4.6

(a) The t statistic for $\beta_1 = 0$ equals $t_0 = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$, which is $t(n-2)$ under $\rho = 0$.

(b) (Fisher) $Z = \frac{1}{2}\log\frac{1+r}{1-r}$ is approximately $N\big(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\big)$ for moderate n (say $n \geq 25$).

Proof. (a) $SSE = SST(1-r^2)$ and $\hat\beta_1^2 S_{xx} = SSR = r^2 SST$, so

$$t_0^2 = \frac{\hat\beta_1^2 S_{xx}}{SSE/(n-2)} = \frac{(n-2)r^2}{1-r^2}.$$

The null distribution is conditionally $t(n-2)$ given the x_i, hence unconditionally as well.

(b) is the variance-stabilizing transformation of the asymptotic law $\sqrt{n}(r-\rho) \to N(0, (1-\rho^2)^2)$; the correction $n-3$ improves the approximation. ■

Part (b) gives tests of $H_0 : \rho = \rho_0 \neq 0$ and confidence intervals: $\tanh\big(Z \pm z_{\alpha/2}/\sqrt{n-3}\big)$ covers ρ with probability about $1-\alpha$.

4.4 Residuals

Motivation. Residuals reveal violations of linearity, constant variance and normality.

Residuals are our window on the errors. In vector form $\mathbf{e} = (I-H)\mathbf{y}$ with H the hat matrix (Chapter 6), and:

Theorem 4.7

$Ee_i = 0$ and

$$\operatorname{Var}(e_i) = \sigma^2\Big(1 - \frac{1}{n} - \frac{(x_i-\bar x)^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(e_i, e_j) = -\sigma^2\Big(\frac{1}{n} + \frac{(x_i-\bar x)(x_j-\bar x)}{S_{xx}}\Big) \ (i \neq j).$$

Proof. $e_i = y_i - \bar y - \hat\beta_1(x_i - \bar x)$. Using $\operatorname{Var}\bar y = \sigma^2/n$, $\operatorname{Var}\hat\beta_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(y_i, \bar y) = \sigma^2/n$, $\operatorname{Cov}(y_i, \hat\beta_1) = \sigma^2 k_i$ and $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$,

$$\operatorname{Var} e_i = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}} - \frac{2}{n} - 2\frac{(x_i-\bar x)^2}{S_{xx}}\Big),$$

which simplifies to the claim; the covariance is computed in the same way. ■

So residuals are dependent with unequal variances (smaller far from $\bar x$; Chapter 10). As $\sum e_i = \sum x_i e_i = 0$, plots against x_i or $\hat y_i$ have no linear trend by construction; look for *non-random* structure:

- a curved band suggests a missing quadratic term or a transformation;
- a funnel suggests nonconstant variance (weighted least squares or a variance-stabilizing transformation of y);
- a trend in time order suggests correlated errors (Chapter 12).

4.5 Selected exercises

Motivation. Independence of the coefficient estimators, the size of the one-sided t test, $E(SSE)$ from residual variances, and design control of $\operatorname{Var}\hat\beta_1$.

Exercise 4.1 (Independence of the coefficient estimators)

Show that $\bar{y}$ and $\hat{\beta}_1$ are always independent, and that $\hat{\beta}_0$ and $\hat{\beta}_1$ are independent if and only if $\bar{x} = 0$.

Solution.

Step 1. (Joint normality.) Each of $\bar{y}$, $\hat{\beta}_0$, $\hat{\beta}_1$ is a fixed linear combination of $\mathbf{y} = (y_1, \dots, y_n)'$, so $(\hat{\beta}_0, \hat{\beta}_1, \bar{y})' = A\mathbf{y}$ for a fixed $3 \times n$ matrix A. Since $\mathbf{y} \sim N_n(X\beta, \sigma^2 I)$, $A\mathbf{y}$ is multivariate normal (Section 5.4).

Step 2. (Uncorrelated implies independent.) For jointly normal (U, W) with Cov $= 0$ the mgf $\exp\{t_1 EU + t_2 EW + \frac{1}{2}(t_1^2 \operatorname{Var} U + t_2^2 \operatorname{Var} W)\}$ factors; conversely independence gives zero covariance.

Step 3. By Theorem 4.1(b), $\operatorname{Cov}(\bar{y}, \hat{\beta}_1) = 0$, so $\bar{y}$ and $\hat{\beta}_1$ are independent for every design.

Step 4. $\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$ (Theorem 4.1(c)) vanishes iff $\bar{x} = 0$, so by Step 2 independence holds iff $\bar{x} = 0$ (then $\hat{\beta}_0 = \bar{y}$). ■

Exercise 4.2 (A one-sided test with a composite null)

Let $0 < \alpha < 1$ and test $H_0 : \beta_1 \le 0$ against $H_1 : \beta_1 > 0$. Show that the test rejecting H_0 when

$$t_0 = \frac{\hat{\beta}_1}{\sqrt{MSE/S_{xx}}} > t_\alpha(n-2)$$

satisfies

$$\sup_{\beta_0 \in \mathbb{R},\ \beta_1 \le 0,\ \sigma^2 > 0} P_{\beta_0, \beta_1, \sigma^2}(\text{the test rejects } H_0) = \alpha,$$

and that its rejection probability exceeds α whenever $\beta_1 > 0$.

Solution.

Step 1. For every parameter value, $T = (\hat{\beta}_1 - \beta_1)/\sqrt{MSE/S_{xx}} \sim t(n-2)$ (Section 4.1). Since $MSE > 0$ with probability one,

$$t_0 = T + \frac{\beta_1}{\sqrt{MSE/S_{xx}}}.$$

Step 2. If $\beta_1 \le 0$, the last term is ≤ 0, so $t_0 \le T$ and $\{t_0 > t_\alpha(n-2)\} \subseteq \{T > t_\alpha(n-2)\}$. Hence the rejection probability is at most $P(T > t_\alpha(n-2)) = \alpha$.

Step 3. At $\beta_1 = 0$, $t_0 = T$ and the rejection probability is exactly α for all β_0, σ^2; with Step 2 the supremum over $\{\beta_1 \le 0\}$ is α, attained at $\beta_1 = 0$.

Step 4. (Power.) $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$ is independent of MSE (Theorem 4.2). Write $c = t_\alpha(n-2)$ and $\tau = \sigma/\sqrt{S_{xx}}$. Conditioning on MSE,

$$P(t_0 > c) = E\Big[P\big(\hat{\beta}_1 > c\sqrt{MSE/S_{xx}} \mid MSE\big)\Big] = E\Big[1 - \Phi\Big(\frac{c\sqrt{MSE/S_{xx}} - \beta_1}{\tau}\Big)\Big].$$

The integrand increases strictly in β_1, so the rejection probability does too: it exceeds α for $\beta_1 > 0$ (unbiased test; this also reproves Step 2). ■

Exercise 4.3 (Expected residual sum of squares without chi-square theory)

(a) Using only Theorem 4.7, show that $E(SSE) = (n-2)\sigma^2$.
(b) Show that $\operatorname{Var} \hat{y}_i + \operatorname{Var} e_i = \sigma^2$ for each i, where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, and that $\sum_i \operatorname{Var} \hat{y}_i = 2\sigma^2$.

Solution [Supplement].

(a) Since $Ee_i = 0$, $Ee_i^2 = \operatorname{Var} e_i$. Summing the formula of Theorem 4.7 and using $\sum(x_i - \bar{x})^2 = S_{xx}$,

$$E(SSE) = \sum_i Ee_i^2 = \sigma^2\Big(n - 1 - \frac{S_{xx}}{S_{xx}}\Big) = (n-2)\sigma^2.$$

No normality is needed: only $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I$ were used.

(b) By Theorem 4.5(a) at $x = x_i$, $\operatorname{Var}\hat{y}_i = \sigma^2\{1/n + (x_i - \bar{x})^2/S_{xx}\}$; adding $\operatorname{Var} e_i$ from Theorem 4.7 gives σ^2. Summing over i,

$$\sum_i \operatorname{Var}\hat{y}_i = \sigma^2\Big(1 + \frac{S_{xx}}{S_{xx}}\Big) = 2\sigma^2.$$

So the total variance $n\sigma^2$ of the y_i splits into $2\sigma^2$ in the fitted values (two parameters) and $(n-2)\sigma^2$ in the residuals. ■

Exercise 4.4 (The best design for the slope)

Let n be even, and suppose the experimenter may choose each x_i in a fixed interval $[a, b]$. Show that

$$\operatorname{Var}\hat{\beta}_1 \geq \frac{4\sigma^2}{n(b-a)^2},$$

with equality if and only if half of the x_i equal a and the other half equal b.

Solution [Supplement].

Step 1. For any constant c, $\sum(x_i - c)^2 = S_{xx} + n(\bar{x} - c)^2 \geq S_{xx}$ (expand $x_i - c = (x_i - \bar{x}) + (\bar{x} - c)$; the cross term vanishes). Take $c = m = (a+b)/2$.

Step 2. Since $x_i \in [a, b]$, $|x_i - m| \leq (b-a)/2$. Hence

$$S_{xx} \leq \sum(x_i - m)^2 \leq \frac{n(b-a)^2}{4}, \qquad \operatorname{Var}\hat{\beta}_1 = \frac{\sigma^2}{S_{xx}} \geq \frac{4\sigma^2}{n(b-a)^2}.$$

Step 3. Equality in Step 2 iff $\bar{x} = m$ and $|x_i - m| = (b-a)/2$, i.e. all $x_i \in \{a, b\}$ with $n_a a + (n - n_a)b = n(a+b)/2$, i.e. $n_a = n/2$; this design gives $S_{xx} = n(b-a)^2/4$ (Exercise 3.2).

It cannot reveal curvature; in practice some interior points check linearity. ■